LESSON PLAN · SCRATCH JUNIOR (K–2) ADVANCED: GAME BUILDERS

Lesson 10: Design a Game Level

Module 4 | 30 minutes | Design and efficiency | KG–Grade 2 · ages 6–7

Objective: Design a fair level with a path, treasure and a goal, then solve it with as few blocks as you can.

Success criteria

- I can solve harder levels.
- I can design a fair level.
- I can use few blocks.

Key vocabulary

- **level**: One stage of a game.
- **designer**: The person who plans the game.
- **fair**: Possible, not too easy, not too hard.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (30 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Obstacle course (5 min): design a 3-obstacle course with cushions.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Design and solve: learners follow the steps.
5 min	**Perform**	Level 1, Level 2, Fair design
3 min	**Probe**	Level 3
4 min	**Wrap-up**	Exit ticket: What makes a level fair?

Unplugged warm-up

Obstacle course (5 min): design a 3-obstacle course with cushions.

Teaching the idea

Design thinking with progressive difficulty; solving with efficient code; ensuring solvability (fairness).

Analogy: Building an obstacle course in the playground.

Live demo: Solve the three levels together; discuss how difficulty rises.

Questions to ask

Ask	Listen for
What makes level 2 harder?	Walls.
Is your level possible?	Test it.
How few blocks can you use?	Count.

Differentiation

Support
Solve levels 1–2 only.

Stretch
Design a level that needs a repeat.

Common misconceptions

Misconception	How to fix it
Harder is always better	Fair is better.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

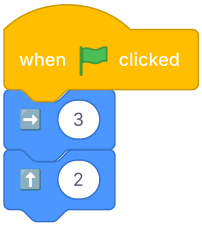
Exit ticket
What makes a level fair?

Homework
Draw a level for someone at home to solve.

Answer key — Kitabash activities

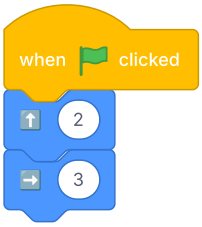
1. Level 1 (perform · block code)

One solution:



2. Level 2 (perform · block code)

One solution:



3. Level 3 (probe · block code)

One solution:



4. Fair design (perform · multiple choice)

Answer: Move a wall so there's a path

Answer key — worksheet

1. **Key words:** level = One stage of a game. · designer = The person who plans the game. · fair = Possible, not too easy, not too hard.
2. **Fair design:** A. Move a wall so there’s a path — A fair level must be possible.
3. **Design a level:** A solvable level.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.